

Presentation: Introduction to General Agent Problem Solving (GAPS)

Slide 1: Title Slide

Introduction to General Agent Problem Solving (GAPS)

- Panaversity Certified AI Development Program
- Lecture 01
- AI-Driven Development, Claude Code, and OpenClaw Fundamentals
- Presented by Sir Zia Khan, Waniya Kazmi, and Ammar

Speaker Notes: Introduce the course, faculty, and the purpose of the lecture.

Slide 2: Learning Objectives

Learning Objectives

- Understand the evolution of AI systems
- Learn the concept of Agentic AI
- Understand the GAPS framework
- Explore self-correcting AI systems
- Learn the importance of Human-in-the-Loop (HITL)

Speaker Notes: Explain what students should learn by the end of the lecture.

Slide 3: Meet the Faculty

Faculty & Leadership Team

- Sir Zia Khan
- Waniya Kazmi
- Sir Ammar
- Experienced educators and AI practitioners
- Dedicated to building world-class AI talent

Speaker Notes: Briefly introduce the faculty and their backgrounds.

Slide 4: Program Vision

Beyond AI Usage

- Move from AI users to AI builders
- Focus on Agentic AI systems
- Develop entrepreneurial thinking
- Build enterprise-grade AI solutions
- Emphasize continuous learning

Speaker Notes: Discuss the vision and long-term goals of the program.

Slide 5: Student Expectations

Code of Conduct

- Avoid procrastination
- Practice consistently
- Build projects continuously
- Participate actively
- Take ownership of learning

Speaker Notes: Highlight the importance of discipline and execution.

Slide 6: The Microwave Analogy

Understanding Intelligence

- Traditional microwave follows fixed settings
- Human acts as sensor and controller
- Smart microwave senses conditions
- Makes decisions automatically
- Adapts based on context

Speaker Notes: Use the microwave example to explain intelligent behavior.

Slide 7: Evolution of AI

AI Paradigm Shift

- Traditional Automation
- Generative AI
- Agentic AI
- Increasing autonomy

- Increasing problem-solving capability

Speaker Notes: Explain how AI evolved over time.

Slide 8: Traditional Automation

Rule-Based Systems

- Uses fixed rules
- Works for predictable scenarios
- Fails on unexpected situations
- Requires manual updates
- Limited adaptability

Speaker Notes: Discuss the limitations of traditional automation.

Slide 9: Generative AI

Prompt Engineering Era

- Human-driven interactions
- Single-turn conversations
- Produces content on demand
- Requires continuous guidance
- Acts as an assistant

Speaker Notes: Explain the strengths and limitations of prompting-based AI.

Slide 10: Agentic AI

Autonomous Systems

- Goal-oriented behavior
- Multi-step reasoning
- Tool usage
- Self-correction
- Independent execution

Speaker Notes: Introduce the concept of AI agents.

Slide 11: What is GAPS?

General Agent Problem Solving

- Framework for intelligent agents
- Supports autonomous decision-making
- Uses perception, reasoning, and action
- Designed for complex tasks
- Foundation of Agentic AI

Speaker Notes: Define GAPS and its purpose.

Slide 12: Perception Layer

Sensors

- Reads files
- Monitors databases
- Consumes APIs
- Observes environments
- Collects required information

Speaker Notes: Explain how agents gather information.

Slide 13: Brain Layer

Reasoning Core

- Powered by Large Language Models
- Goal decomposition
- Planning and reasoning
- Memory management
- Decision making

Speaker Notes: Discuss the intelligence layer of the agent.

Slide 14: Action Layer

Effectors & Tools

- Executes code
- Calls APIs
- Writes files
- Updates databases

- Produces outcomes

Speaker Notes: Explain how agents perform actions.

Slide 15: Goal Decomposition

Breaking Down Complex Problems

- Start with a high-level goal
- Divide into smaller tasks
- Prioritize execution
- Manage dependencies
- Track progress

Speaker Notes: Show how agents convert goals into actionable plans.

Slide 16: Self-Correction Loop

Autonomous Debugging

- Execute task
- Detect errors
- Analyze stack traces
- Rewrite code
- Retry execution

Speaker Notes: Explain how agents learn from failures.

Slide 17: Risks of Full Autonomy

Potential Challenges

- Infinite loops
- Resource exhaustion
- Incorrect outputs
- System corruption
- Unintended actions

Speaker Notes: Discuss the dangers of unrestricted autonomy.

Slide 18: Human-in-the-Loop (HITL)

Safety & Governance

- Human approval checkpoints
- Risk mitigation
- Better accountability
- Controlled execution
- Increased trust

Speaker Notes: Explain why HITL is critical in production systems.

Slide 19: Assignment & Practical Learning

Student Action Plan

- Read assigned GAPS material
- Analyze existing workflows
- Design AI agents
- Share solutions
- Participate in discussions

Speaker Notes: Describe the learning activities expected from students.

Slide 20: Key Takeaways & Conclusion

Key Takeaways

- AI is evolving toward autonomy
- Agentic AI extends beyond prompting
- GAPS provides a structured framework
- Self-correction improves reliability
- HITL ensures safety and control

Speaker Notes: Summarize the lecture and conclude the presentation.